

PANORAMA: Full Proofs

Technical Report

Companion to PANORAMA: *Fast-Track Nearest Neighbors*

Abstract

This technical report gives full derivations for the cost-model propositions in PANORAMA: *Fast-Track Nearest Neighbors*—Propositions 5.1, 5.2, and 5.3—together with the PCA energy-optimality result behind the choice of basis (stated informally in Sections 4.1 and 5.1 of the main paper). Results are numbered to match the main paper; auxiliary lemmas introduced only in this report use a separate “A.” numbering. We collect those auxiliary lemmas (Ky Fan, Dvoretzky–Kiefer–Wolfowitz, and the Gaussian-quantile tail asymptotic) and distinguish high-probability bounds from the calibrated statistical model for the candidate stream.

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1 Notation and assumptions

We adopt the notation of the main paper. All vectors lie in $\mathbb{R}^d$. For an orthogonal transform T and a level $0 \leq m \leq d$, the *tail energy* of a vector $\mathbf{v}$ is

$$R_{T(\mathbf{v})}^{(m,d)} := \sum_{j=m+1}^d T(\mathbf{v})_j^2.$$

The level- m Cauchy–Schwarz lower bound on $\|\mathbf{q} - \mathbf{x}\|^2$ is

$$\text{LB}^m(\mathbf{q}, \mathbf{x}) = R_{T(\mathbf{q})}^{(0,d)} + R_{T(\mathbf{x})}^{(0,d)} - 2 \left(p^{(0,m)}(\mathbf{q}, \mathbf{x}) + \sqrt{R_{T(\mathbf{q})}^{(m,d)} R_{T(\mathbf{x})}^{(m,d)}} \right), \quad (1)$$

where $p^{(0,m)}(\mathbf{q}, \mathbf{x}) = \sum_{j=1}^m T(\mathbf{q})_j T(\mathbf{x})_j$. The ε -relaxed bound replaces the second factor by $2\varepsilon \sqrt{R_{T(\mathbf{q})}^{(m,d)} R_{T(\mathbf{x})}^{(m,d)}}$, with $\varepsilon = 1$ recovering (1).

PANORAMA maintains a running threshold τ , the squared distance of the k^{th} nearest neighbour found so far (written d_k in the main paper). A candidate $\mathbf{x}$ is pruned at level m if $\text{LB}^m(\mathbf{q}, \mathbf{x}) > \tau$, equivalently if the gap

$$\|\mathbf{q} - \mathbf{x}\|^2 - \text{LB}^m(\mathbf{q}, \mathbf{x}) \leq \Delta := \|\mathbf{q} - \mathbf{x}\|^2 - \tau.$$

We call Δ the *pruning margin*. In strict mode ($\varepsilon = 1$), true k -nearest neighbours satisfy $\Delta \leq 0$ and are never pruned by a valid lower bound; the analysis below shows that for non-neighbours Δ is positive with high probability and the gap shrinks fast enough to prune at $m \ll d$.

Asymptotic notation. We use $f(n) \lesssim g(n)$ for model upper-bound scaling after constants are explicit, and $f(n) \sim c g(n)$ for asymptotic equivalence with constant $c > 0$.

Assumptions. The proofs rest on the following assumptions, reproduced here for self-containment.

Assumption A1 (Exponential PCA tail energy). We take T to be the PCA basis of the dataset. On real embeddings its tail energy decays exponentially,

$$R_{T(\mathbf{x})}^{(m,d)} \approx \|\mathbf{x}\|^2 e^{-\alpha m/d},$$

where $\alpha > 0$ is the per-dataset decay rate read directly off the PCA spectrum. This exponential profile is an empirical property of the PCA basis, which we verify directly: the mean absolute fit error of the model is below 0.04 on every dataset (main paper, energy-compaction figure). The $\approx$ records exactly this finite-sample modelling claim.

Assumption A2 (Bounded norms). There is a constant R such that $\|T(\mathbf{x})\| \leq R$ for every $\mathbf{x}$ in the dataset and every query $\mathbf{q}$. We write $C_0 := 4R^2$.

Assumption A3 (Gaussian distance model). For a fixed query $\mathbf{q}$, the squared distances $\|\mathbf{q} - \mathbf{x}\|^2$ in the candidate stream are modelled as i.i.d. draws from a Gaussian with mean μ and standard deviation σ . The exact distribution is chi-square-like and the candidate stream produced by an ANN index is not literally i.i.d.; the Gaussian order-statistic calculation is the tractable cost model whose constants are checked empirically.

Assumption A4 (One dimension per level). The proof reasons at the finest pruning granularity, $L = d$, $m_\ell = \ell$.

A fifth assumption (residual isotropy, A5) is introduced and used only in Section 5.

2 PCA minimises expected tail energy

Lemma A.1 (Ky Fan). *Let $\Sigma \in \mathbb{R}^{d \times d}$ be symmetric positive semidefinite with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d \geq 0$. For every $m \in \{1, \dots, d\}$,*

$$\max_{T \in \text{O}(d)} \sum_{j=1}^m (T \Sigma T^\top)_{jj} = \sum_{j=1}^m \lambda_j,$$

with the maximum attained when the first m rows of T span the eigenspace associated with the top- m eigenvalues of Σ .

Proof (sketch). The diagonal of $T \Sigma T^\top$ is majorised by the eigenvalues of Σ (Schur–Horn theorem). For any m , the partial sum of the m largest diagonal entries is at most the partial sum of the m largest eigenvalues of Σ . Equality is attained by the orthogonal transform whose first m rows span the corresponding eigenspace. See Horn and Johnson [1], Section 4.3 for the full proof. $\square$

Proposition (PCA energy-optimality; cf. main paper Sections 4.1 and 5.1). *Let $X \in \mathbb{R}^d$ be a centred random vector ($\mathbb{E}[X] = \mathbf{0}$) with covariance $\Sigma = \mathbb{E}[XX^\top]$ having eigenvalues $\lambda_1 \geq \dots \geq \lambda_d \geq 0$. For any orthogonal $T \in O(d)$ and any $m \in \{1, \dots, d-1\}$,*

$$\mathbb{E}[R_{T(X)}^{(m,d)}] \geq \sum_{j=m+1}^d \lambda_j,$$

with equality when the rows of T are eigenvectors of Σ ordered by decreasing eigenvalue.

Proof. For any orthogonal T ,

$$\mathbb{E}[(TX)_j^2] = \mathbb{E}[(TX)(TX)^\top]_{jj} = (T \mathbb{E}[XX^\top] T^\top)_{jj} = (T\Sigma T^\top)_{jj}.$$

Summing over $j > m$,

$$\mathbb{E}[R_{T(X)}^{(m,d)}] = \sum_{j=m+1}^d (T\Sigma T^\top)_{jj}.$$

Orthogonal conjugation preserves the trace, so

$$\sum_{j=1}^d (T\Sigma T^\top)_{jj} = \text{Tr}(T\Sigma T^\top) = \text{Tr}(\Sigma) = \sum_{j=1}^d \lambda_j.$$

Therefore

$$\mathbb{E}[R_{T(X)}^{(m,d)}] = \sum_{j=1}^d \lambda_j - \sum_{j=1}^m (T\Sigma T^\top)_{jj}.$$

By Lemma A.1,

$$\sum_{j=1}^m (T\Sigma T^\top)_{jj} \leq \sum_{j=1}^m \lambda_j,$$

with equality when the first m rows of T span the top- m eigenspace of Σ . Substituting,

$$\mathbb{E}[R_{T(X)}^{(m,d)}] \geq \sum_{j=1}^d \lambda_j - \sum_{j=1}^m \lambda_j = \sum_{j=m+1}^d \lambda_j,$$

with equality precisely when T is the PCA basis. $\square$

3 Proposition 5.1: Expected refinement cost

This section derives the expected refinement cost of PANORAMA under the Gaussian candidate-stream model. The argument has three ingredients: a high-probability model for the running k^{th} threshold, a Cauchy–Schwarz pruning-depth bound, and an averaged finite-sample margin constant.

PANORAMA maintains a pruning threshold τ , the squared distance of the k^{th} nearest neighbour found so far. For analytical tractability we model τ_i as the k^{th} order statistic among i i.i.d. draws from the distance distribution, acknowledging that the algorithm’s threshold actually arises from a mixture of exact and pruned candidates.

Lemma A.2 (High-probability bound on the sampled threshold via DKW). *Let the squared distances be i.i.d. with CDF F . For any $\epsilon \in (0, 1)$, with probability at least $1 - 2e^{-2i\epsilon^2}$ (Dvoretzky–Kiefer–Wolfowitz [2]), the k^{th} order statistic τ_i satisfies*

$$F^{-1}\left(\max\left\{0, \frac{k}{i+1} - \epsilon\right\}\right) \leq \tau_i \leq F^{-1}\left(\min\left\{1, \frac{k}{i+1} + \epsilon\right\}\right).$$

Under the Gaussian assumption A3 ($F(r) = \Phi((r - \mu)/\sigma)$), this gives in particular

$$\tau_i \leq \mu + \sigma \Phi^{-1}\left(\frac{k}{i+1} + \epsilon\right) \quad \text{with probability at least } 1 - 2e^{-2i\epsilon^2}.$$

Proof. Let F_i be the empirical CDF of the first i distances. The DKW inequality gives $\Pr(\sup_t |F_i(t) - F(t)| > \epsilon) \leq 2e^{-2i\epsilon^2}$. On the event $\sup_t |F_i - F| \leq \epsilon$ we have $F(t) - \epsilon \leq F_i(t) \leq F(t) + \epsilon$ for all t ; monotonicity of F^{-1} then gives $F^{-1}(u - \epsilon) \leq F_i^{-1}(u) \leq F^{-1}(u + \epsilon)$ for all $u \in (0, 1)$. Taking $u = k/(i + 1)$ and recalling $\tau_i = F_i^{-1}(k/(i + 1))$ yields the two-sided bound. Under A3, $F^{-1}(p) = \mu + \sigma \Phi^{-1}(p)$, which gives the Gaussian form. $\square$

A new candidate is tested against τ_i ; in the Gaussian model, a representative non-near-neighbour candidate has squared distance near the mean μ . This lets us define a model margin.

Definition A.3 (Model margin Δ_i). Fix $\epsilon_i \in (0, 1)$ and set the sampled k^{th} -NN threshold upper bound

$$K_i := \mu + \sigma \Phi^{-1}\left(\frac{k}{i+1} + \epsilon_i\right), \quad \Delta_i := \mu - K_i = -\sigma \Phi^{-1}\left(\frac{k}{i+1} + \epsilon_i\right).$$

On the DKW event of Lemma A.2, the model threshold is at most K_i . Thus a representative non-near-neighbour candidate at squared distance μ has model margin at least Δ_i . We use Δ_i as the per-step representative margin in the cost model; candidate-level fluctuations are absorbed into the averaged constant C_N below. The margin is positive whenever $\frac{k}{i+1} + \epsilon_i < \frac{1}{2}$ (so $\Phi^{-1}(\cdot) < 0$).

Uniform high-probability schedule. Fix a target failure probability $\delta \in (0, 1)$ and set

$$\epsilon_i := \sqrt{\frac{1}{2i} \ln \frac{2N}{\delta}}.$$

By a union bound over $i \in \{k + 1, \dots, N\}$, the event

$$\mathcal{E}_\delta := \bigcap_{i=k+1}^N \left\{ \tau_i \leq \mu + \sigma \Phi^{-1}\left(\frac{k}{i+1} + \epsilon_i\right) \right\}$$

holds with probability at least $1 - \delta$. All bounds below are stated on $\mathcal{E}_\delta$.

A candidate $\mathbf{x}_j$ is pruned at level m once its lower bound exceeds the threshold. A sufficient condition is that the worst-case lower-bound error falls below the margin of the candidate processed at step i :

$$\|\mathbf{q} - \mathbf{x}_j\|^2 - \text{LB}^m(\mathbf{q}, \mathbf{x}_j) < \Delta_i.$$

By Cauchy–Schwarz the left-hand side is at most four times the geometric mean of the tail energies; applying Assumptions A1 and A2,

$$4\sqrt{R_{T(\mathbf{q})}^{(m,d)} R_{T(\mathbf{x}_j)}^{(m,d)}} \leq 4\sqrt{\|\mathbf{q}\|^2 \|\mathbf{x}_j\|^2 e^{-2\alpha m/d}} \leq C_0 e^{-\alpha m/d}, \quad C_0 := 4R^2.$$

Pruning is therefore guaranteed once $C_0 e^{-\alpha m/d} < \Delta_i$, i.e.

$$e^{-\alpha m/d} < \frac{\Delta_i}{C_0} \iff m > \frac{d}{\alpha} \ln \frac{C_0}{\Delta_i}.$$

Lemma A.4 (Model pruning depth). *The modelled number of dimensions ρ_i processed for the candidate at step i is*

$$\rho_i \approx \min\left\{d, \frac{d}{\alpha} \left[\ln \frac{C_0}{\Delta_i}\right]_+\right\}, \quad C_0 = 4R^2, \quad [x]_+ := \max\{0, x\}.$$

The total cost is dominated by the sum of the pruning dimensions over the N candidates. Writing $i_0 := \min\{i \geq k+1 : \frac{k}{i+1} + \epsilon_i < \frac{1}{2}\}$ for the first index at which the model margin becomes positive,

$$\text{Cost} = \sum_{i=k+1}^N \rho_i \approx \sum_{i=\max\{i_0, k+1\}}^N \frac{d}{\alpha} \min\left\{\alpha, \left[\ln \frac{C_0}{\Delta_i}\right]_+\right\}.$$

Let $I_{C_0} := \{i \in \{\max\{i_0, k+1\}, \dots, N\} : \Delta_i \leq C_0\}$ and $N_{C_0} := \max I_{C_0}$. Then

$$\sum_i \left[\ln \frac{C_0}{\Delta_i}\right]_+ = \sum_{i \in I_{C_0}} (\ln C_0 - \ln \Delta_i) = |I_{C_0}| \ln C_0 - \ln \left(\prod_{i \in I_{C_0}} \Delta_i \right).$$

Equivalently, define the finite-sample model constant

$$C_N := \frac{1}{N} \sum_{i=\max\{i_0, k+1\}}^N \min\left\{\alpha, \left[\ln \frac{C_0}{\Delta_i}\right]_+\right\}. \quad (2)$$

This is the average capped log-margin cost per candidate.

Lemma A.5 (Model cost via averaged margins). *The total computational cost is modelled by*

$$\text{Cost} \approx \frac{C_N}{\alpha} Nd.$$

When the cap at d is inactive on the contributing window, this is equivalently

$$C_N \approx \frac{1}{N} \left(|I_{C_0}| \ln C_0 - \ln \prod_{i \in I_{C_0}} \Delta_i \right).$$

The exact finite-sample constant is the average in Lemma A.5. To see its dependence on the Gaussian model parameters, write $p_i := k/(i+1) + \epsilon_i$. When p_i lies in the lower tail, the usual Gaussian-quantile approximation gives

$$\Delta_i = -\sigma \Phi^{-1}(p_i) \approx \sigma \sqrt{2 \ln(1/p_i)}. \quad (3)$$

Substituting (3) into the average shows that the constant has the representative form

$$C_N \approx \frac{|I_{C_0}|}{N} \ln \frac{C_0}{\sigma} - \zeta_N \quad (\text{with the cap at } \alpha \text{ applied candidate-wise}),$$

where

$$\zeta_N := \frac{1}{2N} \sum_{i \in I_{C_0}} \ln(2 \ln(1/p_i)).$$

The term ζ_N varies only through a log-log dependence on the effective tail probabilities. In finite samples we keep C_N as the empirical/model constant rather than treating the endpoint approximation for ζ_N as an exact asymptotic identity.

Using Lemma A.5:

Corollary A.6 (Finite-sample cost form). *On $\mathcal{E}_\delta$, the modelled cost to process the candidate set is*

$$\text{Cost} \lesssim \frac{C_N}{\alpha} Nd, \quad C_N \approx \frac{|I_{C_0}|}{N} \ln \frac{C_0}{\sigma} - \zeta_N.$$

The preceding display should be read as a finite-sample cost model. For a concrete run, the empirical counterpart is

$$\hat{C}_N := \alpha \frac{\text{Cost}}{Nd},$$

the measured constant multiplying $1/\alpha$ in the dimensions-scanned fraction. The Gaussian order-statistic model predicts this constant through the average log margin C_N above. Since the per-candidate work is at most d , the contribution of any failure of $\mathcal{E}_\delta$ is bounded by δNd in expectation:

$$\mathbb{E}[\text{Cost}] \lesssim \mathbb{E}[\text{Cost} \mid \mathcal{E}_\delta] + \delta Nd,$$

so the same C_N/α scaling is unaffected by choosing δ small.

Comparison to the naive cost. The naive, brute-force method computes N full d -dimensional distances, at cost Nd . The bound above is smaller by a factor that scales as α (up to the slowly-varying and logarithmic terms), on the same high-probability event $\mathcal{E}_\delta$.

On the role of α . The parameter α controls the decay $e^{-\alpha m/d}$. If $\alpha \leq 1$, energy decays too slowly (at $m = d/2$ the remaining energy is at least $e^{-1/2}$), leading to weak bounds and limited pruning; effective compaction corresponds to α comfortably greater than 1. The high-probability analysis only replaces the expected-margin terms by their concentrated counterparts and leaves this qualitative conclusion unchanged.

Proposition 5.1 (Expected refinement cost). *Let $\phi := \mathbb{E}[\text{Cost}]/(Nd)$ be the average fraction of dimensions processed per candidate. Under Assumptions A1–A4,*

$$\mathbb{E}[\text{Cost}] \lesssim \frac{C_N}{\alpha} Nd, \quad \phi \lesssim \frac{C_N}{\alpha},$$

where C_N is the finite-sample constant defined in (2). Equivalently, PANORAMA evaluates only a fraction C_N/α of the Nd coordinates a brute-force scan would touch—a speedup of α/C_N over brute force.

Proof. Combining the pruning-depth model with the definition of C_N ,

$$\phi = \frac{\mathbb{E}[\text{Cost}]}{Nd} \approx \frac{1}{\alpha N} \sum_{i=\max\{i_0, k+1\}}^N \min\left\{\alpha, \left[\ln \frac{C_0}{\Delta_i}\right]_+\right\} =: \frac{C_N}{\alpha}.$$

Under the Gaussian margin approximation, this is

$$C_N \approx \frac{|I_{C_0}|}{N} \ln \frac{C_0}{\sigma} - \zeta_N,$$

where ζ_N is the averaged log-log term above. Thus the dependence on N enters through the averaged log-log term. In practice the measured dimensions-scanned fraction reflects the effective constant for the candidate stream; across the datasets in the main paper that constant is order-one and close to 1. $\square$

4 Proposition 5.2: Out-of-distribution cost

In deployment the query $\mathbf{q}$ and the database vectors $\{\mathbf{x}_i\}$ may compact differently under the PCA transform T —the query might come from a different language, sensor, or be adversarial. Let α_q and α_x denote their decay rates:

$$R_{T(\mathbf{q})}^{(m,d)} \approx \|\mathbf{q}\|^2 e^{-\alpha_q m/d}, \quad R_{T(\mathbf{x}_i)}^{(m,d)} \approx \|\mathbf{x}_i\|^2 e^{-\alpha_x m/d}. \quad (4)$$

Proposition 5.2 (Out-of-distribution cost). *With asymmetric compaction rates, the effective decay rate of the lower-bound error is the arithmetic mean $\alpha_{\text{eff}} = (\alpha_q + \alpha_x)/2$, and the expected cost satisfies*

$$\mathbb{E}[\text{Cost}] \lesssim \frac{C_N N d}{\alpha_{\text{eff}}} = \frac{2C_N N d}{\alpha_q + \alpha_x},$$

in the same model sense as Proposition 5.1. If the OOD shift changes only the tail compaction rates, the same slowly varying margin constant C_N applies; if it also changes the distance distribution, the bound holds with a modified margin constant C'_N reflecting that distribution.

Proof. Only the per-candidate depth changes; the threshold, margin, and aggregation arguments are unchanged. From the same Cauchy–Schwarz bound, the lower-bound error is at most $4\sqrt{R_{T(\mathbf{q})}^{(m,d)} R_{T(\mathbf{x}_j)}^{(m,d)}}$. Substituting the asymmetric tails (4) and bounding norms by Assumption A2,

$$4\sqrt{R_{T(\mathbf{q})}^{(m,d)} R_{T(\mathbf{x}_j)}^{(m,d)}} \leq 4R^2 \sqrt{e^{-\alpha_q m/d} e^{-\alpha_x m/d}} = 4R^2 e^{-(\alpha_q + \alpha_x) m/(2d)} = C_0 e^{-\alpha_{\text{eff}} m/d},$$

with $\alpha_{\text{eff}} = (\alpha_q + \alpha_x)/2$. Solving $C_0 e^{-\alpha_{\text{eff}} m/d} < \Delta_i$ gives the capped per-candidate depth $\rho_i \approx \min\{d, (d/\alpha_{\text{eff}})[\ln(C_0/\Delta_i)]_+\}$. The threshold, margin, and aggregation arguments above are unchanged with α replaced by α_{eff} , yielding the stated cost. $\square$

Remark (Graceful degradation). Even a fully out-of-distribution query with no compaction ($\alpha_q \rightarrow 0$) leaves $\mathbb{E}[\text{Cost}] \lesssim 2C_N N d/\alpha_x$: PANORAMASTILL benefits from the database compaction alone, a speedup of roughly $\alpha_x/(2C_N)$ over brute force when the margin constant is unchanged. Performance degrades gracefully rather than collapsing, as long as the database vectors are well-compacted.

5 Proposition 5.3: The ε -relaxation

Setting $\varepsilon = 1$ uses the exact Cauchy–Schwarz correction $2C^m$ and preserves recall. That worst-case correction is attained only at the adversarial residual alignment $\rho_m = -1$; on real, isotropised residuals the typical alignment is $\rho_m \approx 0$, where the correction is already a factor of two smaller. The ε -relaxation exploits this gap: it discounts the correction by a factor $\varepsilon \in (0, 1]$, pruning earlier at a controlled, quantifiable cost in recall. We first record the exact gap identity, then state the isotropy assumption that makes the typical case precise, and finally turn it into a cost bound that mirrors Proposition 5.1.

The ε -relaxed bound is

$$\text{LB}_\varepsilon^m(\mathbf{q}, \mathbf{x}) = A^m - 2\varepsilon C^m, \quad C^m := \sqrt{R_{T(\mathbf{q})}^{(m,d)} R_{T(\mathbf{x})}^{(m,d)}},$$

where $A^m = R_{T(\mathbf{q})}^{(0,d)} + R_{T(\mathbf{x})}^{(0,d)} - 2p^{(0,m)}(\mathbf{q}, \mathbf{x})$. Define the residual inner product

$$\pi_m = \sum_{j>m} T(\mathbf{q})_j T(\mathbf{x})_j$$

and its normalised value

$$\rho_m := \frac{\pi_m}{C^m} \in [-1, 1],$$

where the bounds follow from Cauchy–Schwarz applied to the residual.

Lemma A.7 (Gap identity). *For every level m ,*

$$\|\mathbf{q} - \mathbf{x}\|^2 - \text{LB}_\varepsilon^m(\mathbf{q}, \mathbf{x}) = 2C^m(\varepsilon - \rho_m).$$

Proof. Direct expansion:

$$\begin{aligned}\|\mathbf{q} - \mathbf{x}\|^2 &= R_{T(\mathbf{q})}^{(0,d)} + R_{T(\mathbf{x})}^{(0,d)} - 2 \sum_{j=1}^d T(\mathbf{q})_j T(\mathbf{x})_j \\ &= A^m - 2\pi_m.\end{aligned}$$

Subtracting the relaxed bound,

$$\|\mathbf{q} - \mathbf{x}\|^2 - \text{LB}_\varepsilon^m(\mathbf{q}, \mathbf{x}) = (A^m - 2\pi_m) - (A^m - 2\varepsilon C^m) = 2(\varepsilon C^m - \pi_m) = 2C^m(\varepsilon - \rho_m). \quad \square$$

At $\varepsilon = 1$ and worst case $\rho_m = -1$ the gap is $4C^m$, matching the strict cost bound; at $\varepsilon = 1$ and typical $\rho_m \approx 0$ the gap is $2C^m$, a factor of 2 tighter.

Assumption A5 (Residual isotropy). For non-near-neighbour pairs $(\mathbf{q}, \mathbf{x})$, the residual coordinates $(T(\mathbf{q})_j, T(\mathbf{x})_j)_{j>m}$ are independent and uniformly distributed on the sphere in $\mathbb{R}^{d-m}$ (independently for $\mathbf{q}$ and $\mathbf{x}$). Consequently, by standard high-dimensional concentration,

$$\rho_m \sim \mathcal{N}\left(0, \frac{1}{d-m}\right) \quad \text{approximately.}$$

Remark. Assumption A5 is realised by the intra-level isotropization step in the main paper’s variance-shaping transform. It applies only to non-near-neighbour pairs; for true k -NN pairs the residuals are correlated (the nearness manifests in the residual), so ρ_m for true neighbours concentrates strictly above 0. The recall loss at $\varepsilon < 1$ arises precisely from neighbour pairs with $\rho_m > \varepsilon$.

Lemma A.8 (Pruning depth, ε -relaxed). *Under Assumptions A1–A5, for non-near-neighbour candidates and a margin $\Delta > 0$, the pruning depth at the ε -relaxed bound is*

$$d^{(\varepsilon)}(\Delta) \approx \frac{d}{\alpha} \ln \frac{2R^2}{\Delta} + \frac{d}{\alpha} \ln \varepsilon.$$

Proof. Under A5, ρ_m concentrates at 0 with deviation $1/\sqrt{d-m}$. Substituting $\rho_m \approx 0$ in Lemma A.7,

$$\|\mathbf{q} - \mathbf{x}\|^2 - \text{LB}_\varepsilon^m(\mathbf{q}, \mathbf{x}) \approx 2\varepsilon C^m.$$

Applying Assumption A1 to both factors of C^m and Assumption A2,

$$2\varepsilon C^m \approx 2\varepsilon \|\mathbf{q}\| \|\mathbf{x}\| e^{-\alpha m/d} \leq 2\varepsilon R^2 e^{-\alpha m/d}.$$

Pruning occurs at the smallest m for which this is at most Δ :

$$2\varepsilon R^2 e^{-\alpha m/d} \leq \Delta \iff m \geq \frac{d}{\alpha} \ln \frac{2\varepsilon R^2}{\Delta} = \frac{d}{\alpha} \ln \frac{2R^2}{\Delta} + \frac{d}{\alpha} \ln \varepsilon. \quad \square$$

The first term is the typical-case pruning depth at $\varepsilon = 1$, with prefactor $2R^2$ rather than the strict-mode prefactor $C_0 = 4R^2$. The two prefactors differ by a constant inside the log and are absorbed into the same slowly-varying constant as in Proposition 5.1.

Proposition 5.3 (Cost under ε -relaxation). *Under Assumptions A1–A5 and $\varepsilon \in (1/e, 1]$, the expected refinement cost of PANORAMA with the ε -relaxed bound satisfies*

$$\mathbb{E}[\text{Cost}_\varepsilon] \lesssim \frac{Nd(c_{\text{typ}} + \ln \varepsilon)}{\alpha},$$

where c_{typ} is the typical-residual version of the constant C_N (with prefactor $2R^2$ instead of the strict worst-case $4R^2$). Since the empirical typical-case constant is close to 1, the simplified model is $\mathbb{E}[\text{Cost}_\varepsilon] \lesssim Nd(1 + \ln \varepsilon)/\alpha$, equivalently $\alpha_{\text{eff}}(\varepsilon) \approx \alpha/(1 + \ln \varepsilon)$. For $\varepsilon < 1$, each candidate is pruned $(d/\alpha) \ln(1/\varepsilon)$ dimensions earlier on average than at $\varepsilon = 1$ until the cap at zero dimensions becomes active.

Proof. The argument mirrors the cost proof, with the pruning depth of Lemma A.4 replaced by Lemma A.8. We work on the same high-probability event $\mathcal{E}_\delta$ (Lemma A.2); the threshold and margin analysis are unchanged. Substituting the margin asymptotic (3) into Lemma A.8 and aggregating as in Proposition 5.1, the additive $(d/\alpha) \ln \varepsilon$ term carries through the sum to give

$$\mathbb{E}[\text{Cost}_\varepsilon] \lesssim \frac{Nd}{\alpha} (c_{\text{typ}} + \ln \varepsilon),$$

where $c_{\text{typ}} = \ln(2R^2/\sigma) - \zeta_N$ is the typical-residual counterpart of C_N , shifted only by $\ln 2$ (the typical-case prefactor is $2R^2$ rather than $C_0 = 4R^2$) and inheriting the same log-log behaviour. Since c_{typ} is observed to be ≈ 1 ,

$$\mathbb{E}[\text{Cost}_\varepsilon] \lesssim \frac{Nd(1 + \ln \varepsilon)}{\alpha}. \quad \square$$

Remark (Validity range). The asymptotic requires $c_{\text{typ}} + \ln \varepsilon > 0$. Setting $c_{\text{typ}} = 1$ gives the conventional threshold $\varepsilon > 1/e \approx 0.37$; the actual threshold is dataset-dependent. Below it, the formula predicts a negative pruning depth—an artifact: the bound clears the margin already at $m = 0$, and the cost is governed by other factors (heap warm-up, fixed per-candidate overhead). In this regime ε should be tuned empirically.

Remark (Additive vs. multiplicative). The ε -relaxation contributes *additively* to the per-candidate pruning depth, in contrast to the multiplicative $1/\alpha$ scaling of the base model. A moderate $\varepsilon = 0.7$ shifts the prefactor by $\ln 0.7 \approx -0.36$, enough to produce a noticeable QPS gain without changing the leading $1/\alpha$ behaviour, while preserving near-perfect recall.

Proposition 5.3 characterises the speed side of the ε knob. Pruning at $\varepsilon < 1$ discards true k -nearest-neighbour pairs whose residual ratio ρ_m exceeds ε . The recall loss thus depends on the dataset-specific distribution of ρ_m for true k -NN pairs, which is measurable per dataset (see the ε ablation in the main paper). We do not attempt a closed-form recall bound; the ρ_m distribution for k -NN pairs concentrates strictly above 0 (these pairs are correlated, breaking the isotropy of A5), and recall begins to degrade once ε falls below the bulk of this distribution.

References

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- [2] P. Massart. The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality. *The Annals of Probability*, 18(3):1269–1283, 1990.